

2201 321

Veterinary Virology



Family *Poxviridae*, *Asfarviridae*, *Papillomaviridae*, *Polyomaviridae*, *Iridoviridae*

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วัตถุประสงค์

- นิสิตมีความรู้ความเข้าใจเกี่ยวกับไวรัสและโรคที่เกิดจากไวรัสใน Family *Poxviridae*, *Asfarviridae*, *Papillomaviridae*, *Polyomaviridae* และ *Iridoviridae* ดังนี้
 - ลักษณะและชีววิทยาของไวรัส
 - การก่อโรคในสัตว์ พยาธิกำเนิดและอาการแสดงทางคลินิก
 - ตัวอย่างเพื่อชั้นสูตรทางห้องปฏิบัติการ วิธีการและเทคนิคที่ใช้ชั้นสูตรทางห้องปฏิบัติการ
 - การควบคุมและการป้องกันโรค

Family *Poxviridae*

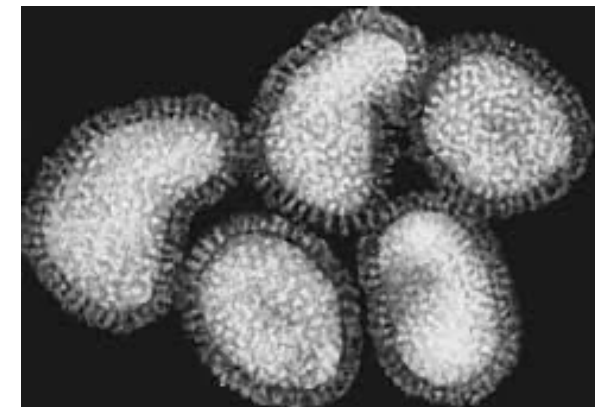
The family includes two subfamilies

- Subfamily *Chordopoxvirinae* infect vertebrates and are classified in the genera;

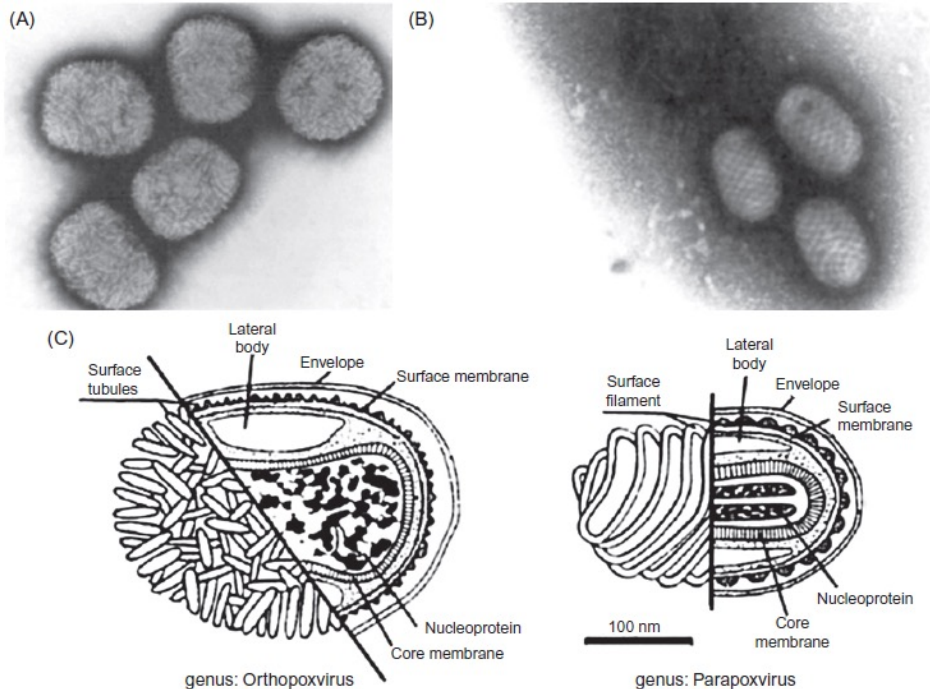
- | | |
|-----------------------------------|-------------------------------------|
| • Genus <i>Avipoxvirus</i> * | • Genus <i>Orthopoxvirus</i> * |
| • Genus <i>Capripoxvirus</i> * | • Genus <i>Oryzopoxvirus</i> |
| • Genus <i>Centapoxvirus</i> | • Genus <i>Parapoxvirus</i> * |
| • Genus <i>Cervidpoxvirus</i> | • Genus <i>Pteropoxvirus</i> |
| • Genus <i>Crocodylidpoxvirus</i> | • Genus <i>Salmonpoxvirus</i> |
| • Genus <i>Leporipoxvirus</i> | • Genus <i>Sciuripoxvirus</i> |
| • Genus <i>Macropopoxvirus</i> | • Genus <i>Suipoxvirus</i> * |
| • Genus <i>Molluscipoxvirus</i> | • Genus <i>Vespertilionpoxvirus</i> |
| • Genus <i>Mustelpoxvirus</i> | • Genus <i>Yatapoxvirus</i> |

- Subfamily *Entomopoxvirinae* infect insects, and are classified into four genera;

- Genus *Alphaentomopoxvirus*
- Genus *Betaentomopoxvirus*
- Genus *Deltaentomopoxvirus*
- Genus *Gammaentomopoxvirus*



Viral properties



From: Fenner, 2011

- Pleomorphic, generally brick-shaped (220–450 nm long, 140–260 nm wide, 140–260 nm thick) with a lipoprotein surface membrane displaying tubular or globular units (10–40 nm)
 - Virions can also be ovoid (250–300 nm long, 160–190 nm diameter) with a surface membrane possessing a regular spiral filament (10–20 nm in diameter)
- Complex structure;
 - Negatively-stained electron microscopy images show that the surface membrane encloses a biconcave or cylindrical core that contains the genome DNA and proteins organized in a nucleoprotein complex
 - One or two lateral bodies appear to be present in the concave region between the core wall and a membrane

Viral properties (cont.)



Genome; a single linear molecule of dsDNA of 128–375 kbp with covalently-closed termini



Poxvirus proteins constitute about 90% of the particle weight

- Poxvirus genomes encode 120–300 proteins depending on the virus; about 100 proteins are present in virions
- Virus particles contain many enzymes involved in DNA transcription or modification of proteins or nucleic acids



Enveloped virions have virus-encoded polypeptides in the lipid bilayer, which surrounds the particle

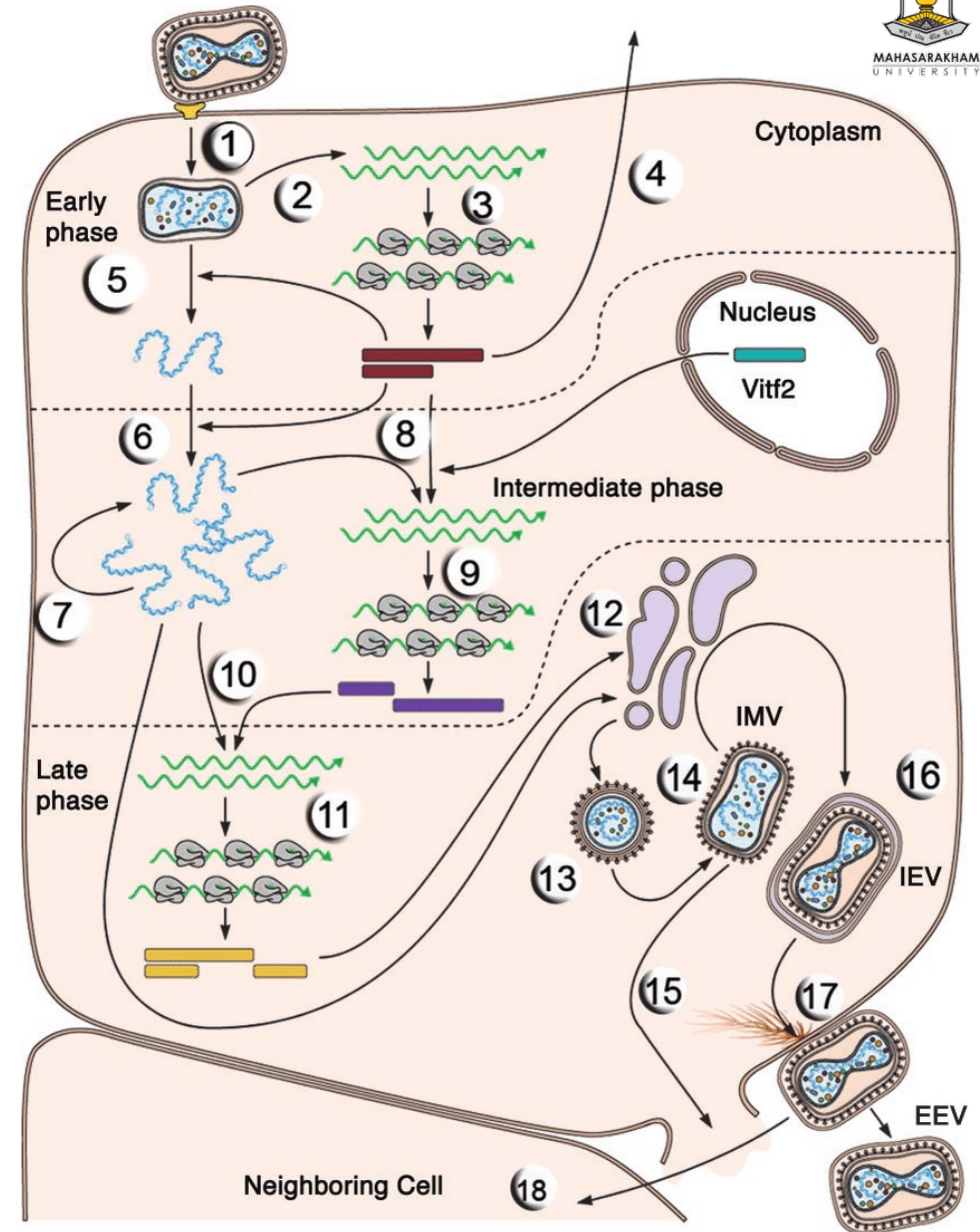


Generally, virion infectivity is sensitive to common detergents, formaldehyde, oxidizing agents and temperatures greater than 40 °C

- Infectivity of some members is resistant to trypsin, but in some cases, it is enhanced by trypsin treatment
- Some members are insensitive to ether
- Virions are relatively stable in dry conditions at room temperature; they can be lyophilized with little loss of infectivity

Viral replication

- Cytoplasm replication
- Virion มีการเชื่อม (fusion) กับ plasma membrane หรือเกิด endocytosis
- Viral core ถูกปล่อยเข้าไปใน cytoplasm
- Transcription สร้าง mRNA และสังเคราะห์โปรตีน, สร้าง DNA (+/-)
- ประกอบเป็น virion ที่สมบูรณ์
- อนุภาคที่สมบูรณ์เคลื่อนไปยัง Golgi complex ได้ envelop
- ออกจากเซลล์โดยกระบวนการ cell lysis



Poxviruses infection and diagnosis

- **Routes of transmission**

- Droplet/aerosol
- Direct contact
- Arthropods (via mechanical means)
- Indirect contact via fomites

- Many poxviruses of vertebrates produce dermal maculopapular, vesicular rashes after systemic or localized infections
 - Poxviruses infecting humans are zoonotic
- Neutralizing antibodies and cell-mediated immunity play a major role in clearance of some vertebrate poxvirus infections
 - Reinfection rates are generally low and usually less severe

- **Diagnosis**

- Clinical signs and lesions
- EM
- Inoculation virus into cell culture
- Pock lesion on chorioallantoic membrane (CAM)





Genus *Avipoxvirus*

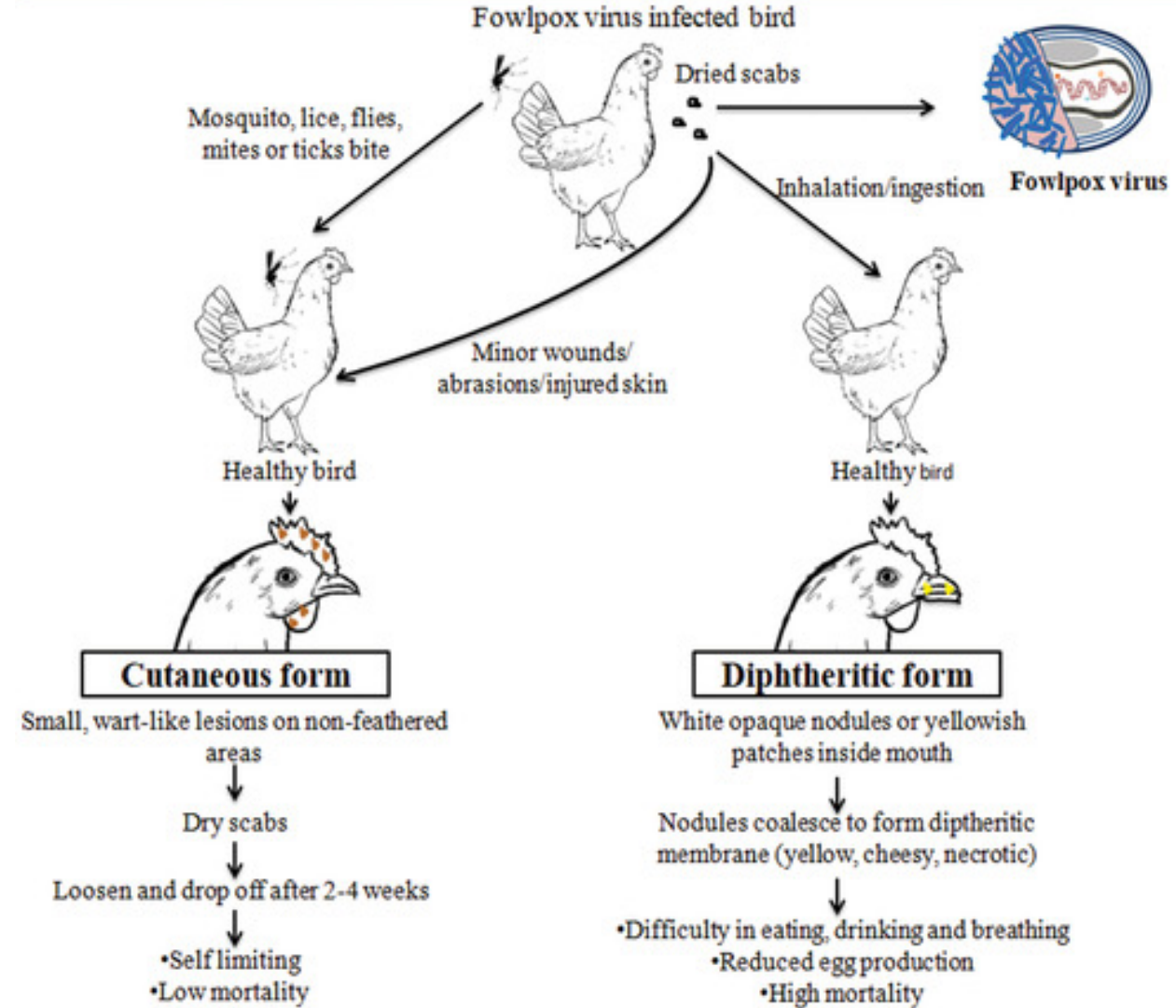
- Pox-like diseases have been described worldwide in over 278 species of birds, spread across 23 orders
- Two forms of disease are commonly found, cutaneous and respiratory/enteric, dependent on the site of infection
 - The cutaneous form tends to be less serious than the respiratory/enteric form with mortality ranging from the low percentages up to 80–100% in some affected flocks
- In commercial poultry flocks, poxvirus infection can result in egg drop, impaired fertility and slower growth rates
- In wild birds increased susceptibility to secondary bacterial infections, impaired vision and ability to feed at will results in greater predation
- Viruses are primarily transmitted mechanically by arthropods, by direct contact with fomites or through aerosols
- Avipoxviruses grow productively in avian cell cultures, but abortively in mammals and mammalian cell lines

Fowlpox virus

- Chicken and turkey; high infection
- **Clinical signs and transmissions**
 - Cutaneous or dry form: *biting arthropods*
 - Papules (comb, wattles, around beak and cloaca, legs, skin around the nares, and eyelids)
 - Healing within 3 wks.
 - Diphtheritic or wet form: *aerosol infection*
 - Necrotic pseudo-membrane (mouth, larynx, pharynx, and trachea)
 - Death by asphyxiation



Transmission cycle



Genus

Capripoxvirus

- Members of the genus *Capripoxvirus* cause economically important, notifiable diseases in domesticated cattle, sheep and goats
 - In cattle, milk drop, secondary mastitis, abortion and infertility can occur, although mortality is usually <10%
 - In lambs and young goats are highly susceptible to infection and mortality can be up to 50%
- Lumpy skin disease virus distribution is confined mainly to Africa and the Middle East, both goatpox virus and sheeppox virus have a much wider distribution across Africa and Asia
- Viruses can be transmitted mechanically by arthropods and by direct contact or fomites
- There is extensive DNA cross-hybridization between species, and serological cross-reaction and cross-protection between the viruses
- Sheeppox and goatpox viruses have a narrow host range, whereas Lumpy skin disease virus has been found in a number of wildlife species across Africa

Sheep pox virus and Goat pox virus

- Sheeppox and goatpox are serious, often fatal, diseases of sheep and goats characterized by widespread skin eruption and respiratory disease
- Incubation period: 8-14 days
- Clinical signs
 - Fever, high respiratory rate, edema of the eyelids, mucous nasal discharge
 - Widespread skin lesions develop and are most evident on the muzzle, ears, and other areas free of wool or long hair
 - In severe cases of sheeppox or goatpox, lesions can develop in the lungs
 - In some sheep, however, the disease is mild or inapparent



Diagnosis and control



- **Diagnosis**

- Diagnosis of sheeppox and goatpox is usually based on PCR assay; however, antigen-capture ELISA, along with characterization of the viruses by culture and genome sequencing, has also been used

- **Control**

- Live attenuated virus vaccines against sheeppox and goatpox are commercially available and induce good immunity and protection
 - However, concerns over reversion to virulence and vaccine spread mean that they are used only in areas where the disease is considered endemic
- Emergency control: rapid identification, isolation and culling, and prevention via movement restrictions and strict biosecurity

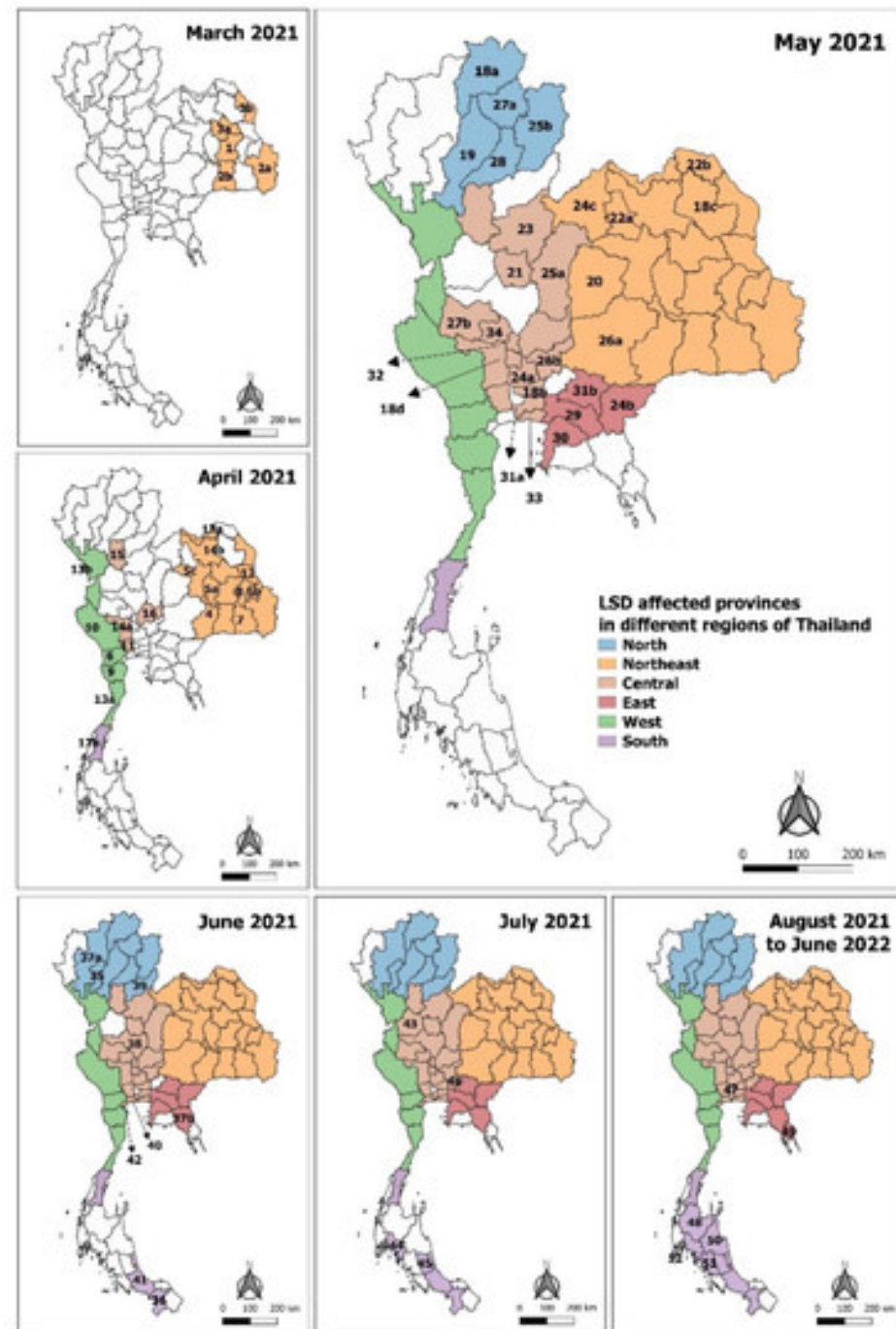


Lumpy skin disease

- Host; Cattle and cape buffalo
- Clinical signs; fever, lacrimation, nasal discharge, loss of appetite, generalized lymphadenitis, edema of the limbs
 - skin nodule (dermis and epidermis) → ulcer → necrosis
 - อัตราการเกิดโรคสูง 100% แต่อัตราการตายต่ำ 1-2% โรคจึงคล้ายคลึงกับ FMD
- การแพร่กระจายในไทย: เดือนมีนาคม พ.ศ.2564 มีรายงานพบโรคลัมปี สกิน ในประเทศไทยเป็นครั้งแรก โดยพบโคมีอาการแสดงคล้ายโรคลัมปี สกิน ซึ่งเป็นโคของเกษตรกรรายย่อยในอ.พนมไพร จ.ร้อยเอ็ด และได้รับการยืนยันผลตรวจทางห้องปฏิบัติการ จากสถาบันสุขภาพสัตว์แห่งชาติ พบว่าตัวอย่างที่มาจากโคป่วยตรวจพบเชื้อไวรัสโรคลัมปี สกิน หลังจากนั้นโรคก็ได้แพร่กระจายไปอีกหลายจังหวัดอย่างรวดเร็วมาจนถึงปัจจุบัน

Distribution of LSD outbreaks in Thailand from March 2021 to June 2022

Suwankitwat N, Songkasupa T, Boonpornprasert P, Sripipattanakul P, Theerawatanasirikul S, Deemagarn T, Suwannaboon M, Arjkumpa O, Buamithup N, Hongsawat A, et al. Rapid Spread and Genetic Characterisation of a Recently Emerged Recombinant Lumpy Skin Disease Virus in Thailand. *Veterinary Sciences*. 2022; 9(10):542. <https://doi.org/10.3390/vetsci9100542>



การใช้วัคซีนป้องกัน โรคลัมปี สกิน



ขนาด 20 มิลลิลิตร (10 โดส)

วัคซีนโรคลัมปี สกิน เป็นวัคซีนเชื้อตาย สื่อน้ำมัน
(Inactivated oil-emulsion vaccine)

ผลิตจากเชื้อไวรัสลัมปี สกิน ซึ่งเตรียมจากเซลล์เพาะเลี้ยง และทำให้อนุฤทธิ์
ด้วยสาร Binary ethylen imine (BEI)

สรรพคุณ

ใช้ฉีดป้องกันโรคลัมปี สกิน
ในโคและกระบือ



วิธีใช้

ฉีดวัคซีนครั้งแรกให้สัตว์อายุมากกว่า 4 เดือนขึ้นไป และฉีดวัคซีนครั้งที่ 2
หลังจากฉีดครั้งแรก 3-4 สัปดาห์ จากนั้นให้ฉีดวัคซีนซ้ำทุก 6 เดือน

กรณีสัตว์ที่เคยได้รับวัคซีนมาก่อน ให้ฉีดวัคซีนเข็มเดียว (ไม่ต้องฉีด
เข็มกระตุ้น) และให้ฉีดวัคซีนซ้ำทุก 6 เดือน

สัตว์อายุ 4 เดือนขึ้นไป



สำนักเทคโนโลยีชีวภัณฑ์สัตว์ กรมปศุสัตว์
กระทรวงเกษตรและสหกรณ์

LSD immunization

• ข้อมูลสำคัญของวัคซีน LSD ในไทย

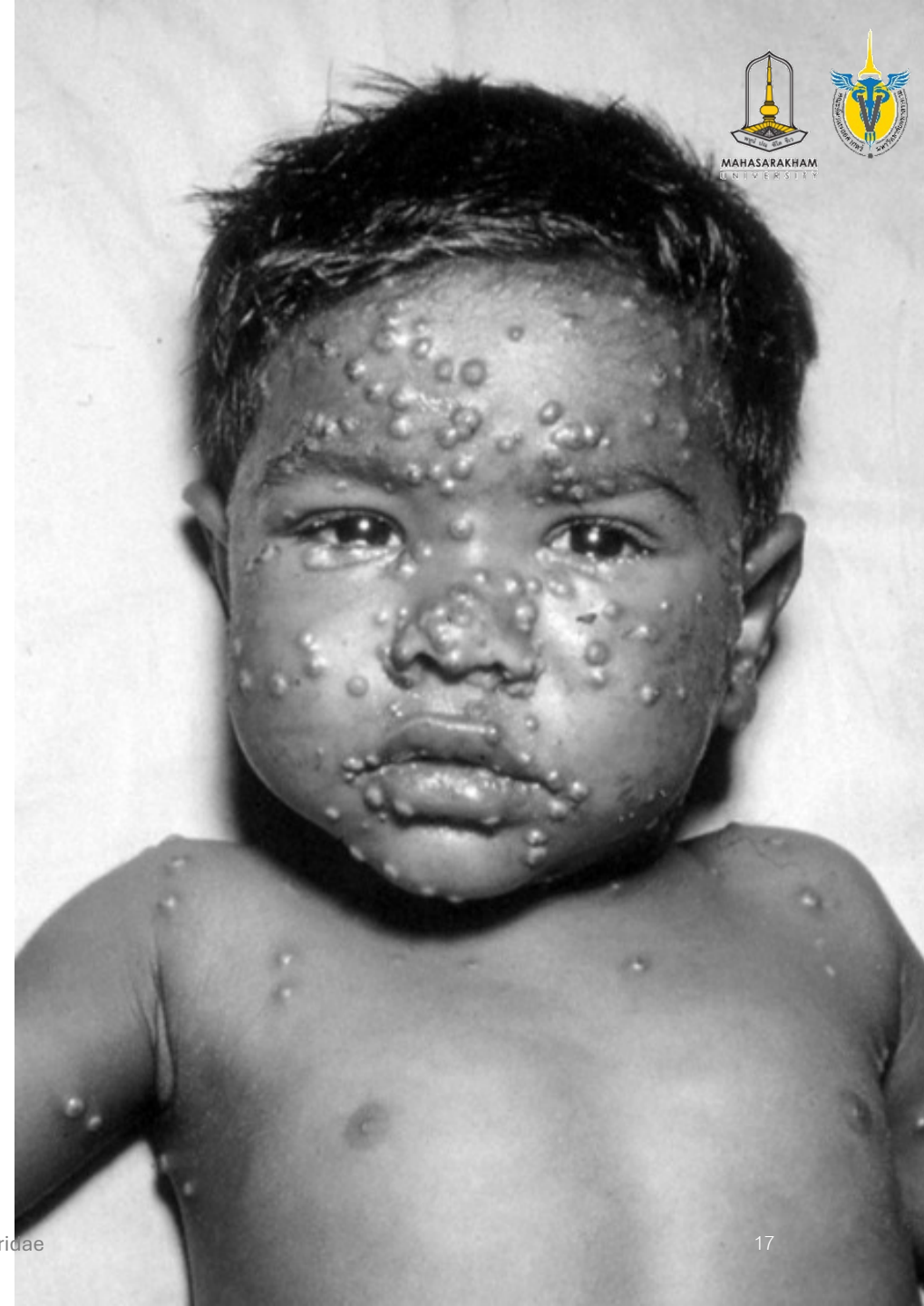
- **ชนิดของวัคซีน:** วัคซีนเชื้อตาย (ทำลายฤทธิ์ด้วยสาร BEI) ซึ่งปลอดภัยต่อสัตว์ ไม่ทำให้เกิดการแพร่กระจายของโรค
- **ข้อห้าม:** ห้ามใช้วัคซีนที่ยังไม่ผ่านการรับรองจากสำนักงานคณะกรรมการอาหารและยา (อย.) เนื่องจากมีความเสี่ยงสูงที่จะนำเชื้อเป็นมาระบาดภายในฝูง

• โปรแกรมการฉีดวัคซีน

- **ช่วงอายุที่ฉีด:** ลูกโคหรือกระบือที่อายุ 4-6 เดือนขึ้นไป
- **ระยะเวลาของภูมิคุ้มกันโรค:** ประมาณ 6 เดือน (ต้องฉีดกระตุ้นซ้ำทุก 6 เดือน เพื่อคงระดับภูมิคุ้มกันในฝูงสัตว์)
- **การจัดการฝูง:** หากพบสัตว์ในฝูงแสดงอาการป่วยแล้ว ให้แยกกักและทำการรักษาตามอาการควบคู่ไปกับการจัดการความปลอดภัยทางชีวภาพ (เช่น การกำจัดแมลงดูดเลือดที่เป็นพาหะ)

Genus *Orthopoxvirus*

- The Orthopoxvirus genus includes 13 species, and divide into two sister clades; 1) the Eurasian/African orthopoxviruses and 2) the North American orthopoxviruses
- The host range varies between species and can be broad in laboratory animals and tissue culture, but relatively narrow in nature
 - For example, members of the species Cowpox virus and Vaccinia virus can infect multiple host species, whereas members of the species Variola virus and Ectromelia virus only infect humans and mice respectively
- Most infections are generalized and disseminated and can be fatal



Cowpox virus



- Cowpox is a rare, self-limiting zoonotic disease
- Reservoir hosts: rodents
- Hosts: cows, humans, zoo animals and domestic cats
- Incubation period: about 5 days
- **Clinical signs**
 - Cattle: pustular lesions (teats and udders)
 - Cats: necrotizing dermatitis and pulmonary infection
 - Humans: maculopapular eruptions

Monkeypox virus

- Monkeypox spreads through close, personal skin-to-skin contact, as well as through contact with respiratory droplets or contaminated items
- **Clinical signs**
- The illness typically lasts 2 to 4 weeks and symptoms usually develop within a week of exposure
 - Common signs include:
 - Rash: Often located on the face, hands, feet, mouth, genitals, or anus
 - The sores develop into raised bumps, fill with fluid or pus, and eventually scab over
 - Flu-like symptoms: Fever, chills, intense headache, muscle aches, back pain, and fatigue
 - Swollen lymph nodes: A distinguishing feature of mpox compared to similar diseases like chickenpox



Genus

Parapoxvirus

- Generally, infect ungulates with disease apparent in livestock species (sheep, cattle, goats) and wildlife (camels)
 - Disease is found world-wide wherever livestock species are farmed
 - Infection does not generate sterile immunity, and animals can be re-infected with the virus albeit resulting in milder disease
- Normally, manifests as a localized erythematous pustular dermatitis
- Although individual parapoxvirus species have a very narrow host range, all are considered zoonotic or potentially zoonotic
- Member species;
 - Orf virus
 - Pseudocowpox virus
 - Bovine papular stomatitis virus



FIGURE 7.9 Orf lesion on the lip of a lamb. (Courtesy K. Thompson, Massey University.)

Orf virus

- A zoonotic parapoxvirus that primarily causes "scabby mouth" or contagious ecthyma in sheep and goats, but can be transmitted to humans
- Natural hosts: sheep and goats
- Lesions: papules, pustules, thick crusts
- Locations: muzzle, lips, gum, tongue, eyelids, feet, and teats



Facial lesions

Lip lesions

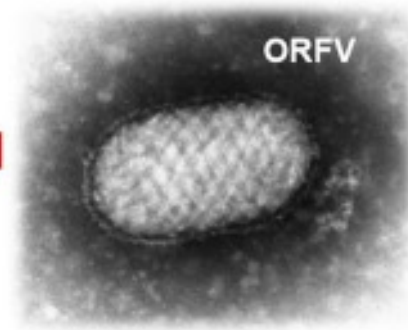
Pinnal lesions

Orbital lesions

Mastitis in ewes

Genital lesions

Footrot



Pseudocowpox virus

- Pseudocowpox is a worldwide, contagious parapoxvirus affecting cattle
- Transmissions
 - Cross-sulking
 - Milking machines
 - Arthropods
- Mild skin lesions on cow teats and udders, typically appearing as swollen orange papules that form characteristic horseshoe-shaped scabs
- Pseudocowpox can spread to humans via direct contact, resulting in a localized skin condition known as "milker's nodules"





Genus *Suipoxvirus*

- **Swinepox virus**

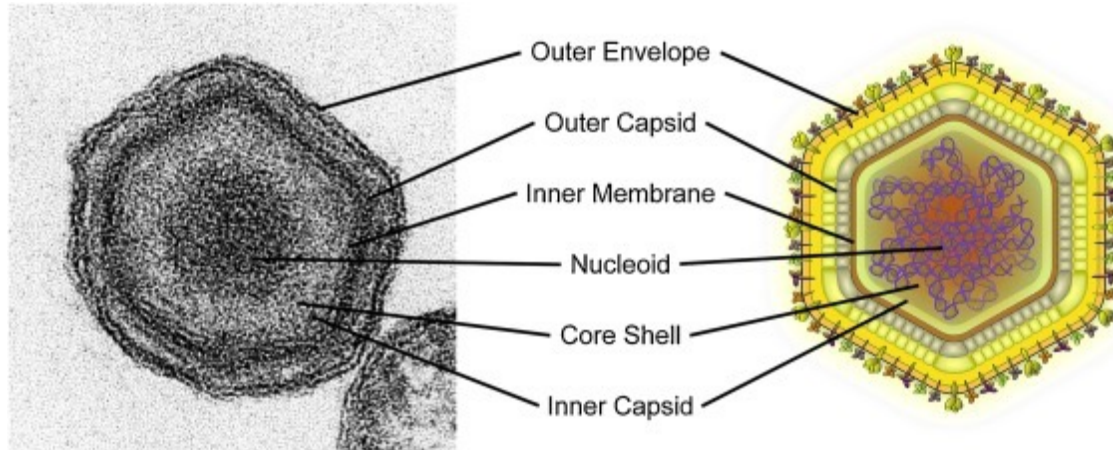
- Worldwide distribution and causes a relatively mild, self-limiting, skin disease in adult pigs with lesions normally confined to the point of infection
- Disease can be more serious in young piglets where morbidity can reach 100%
- Transmission: Pig louse (*Hematopinus suis*), direct contact
- Lesion: Skin (Papules → vesicles → pustules → crust)
- No vaccine
- No evidence that swinepox virus is zoonotic

Family *Asfarviridae*

- African swine fever virus is a large enveloped DNA virus that is the sole member of the genus *Asfivirus* within the family *Asfarviridae*
- **Asfar = African swine fever and related viruses**
- Virus strains are distinguished by their virulence to swine, which ranges from highly lethal to subclinical disease



Virion properties



- Enveloped virus, 200 nm in diameter
- Complex icosahedral capsid
- Genome: linear double-stranded DNA, 170 -190 kbp in size, depending on the virus strain
- Replicate in cytoplasm and nucleus

Epidemiology

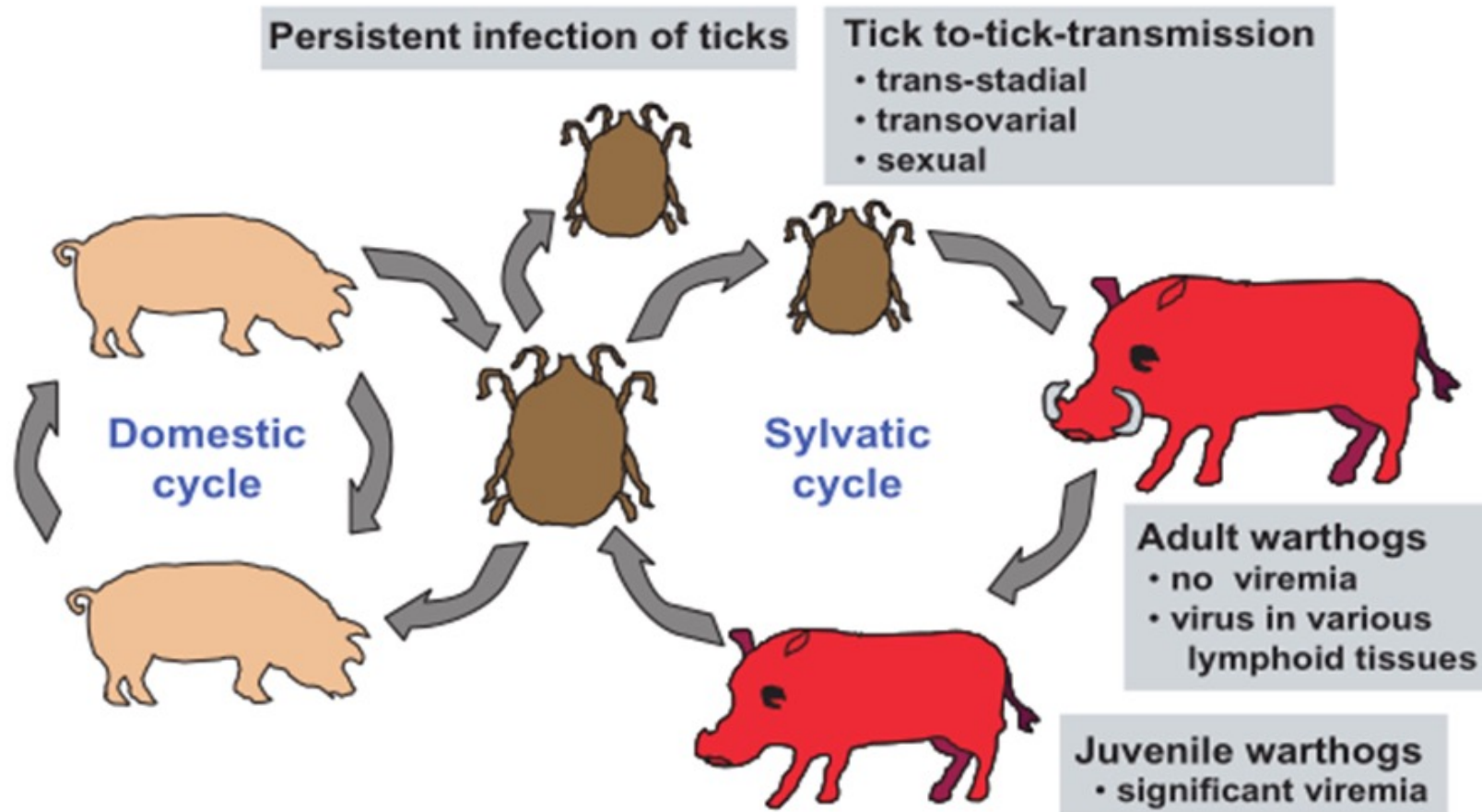
- Spread throughout portions of Georgia, Armenia, Azerbaijan, and Russia
- Enzootic in sub-Saharan Africa and Sardinia
- The presence of wild boar and extensive pig farming sustain enzootic African swine fever in Sardinia
- Infect:
 - Domestic swine and other members of the family *Suidae*, including Warthogs (*Potamochoerus aethiopicus*), Bush pigs (*P. porcus*) and Wild boar (*Sus scrofa ferus*)



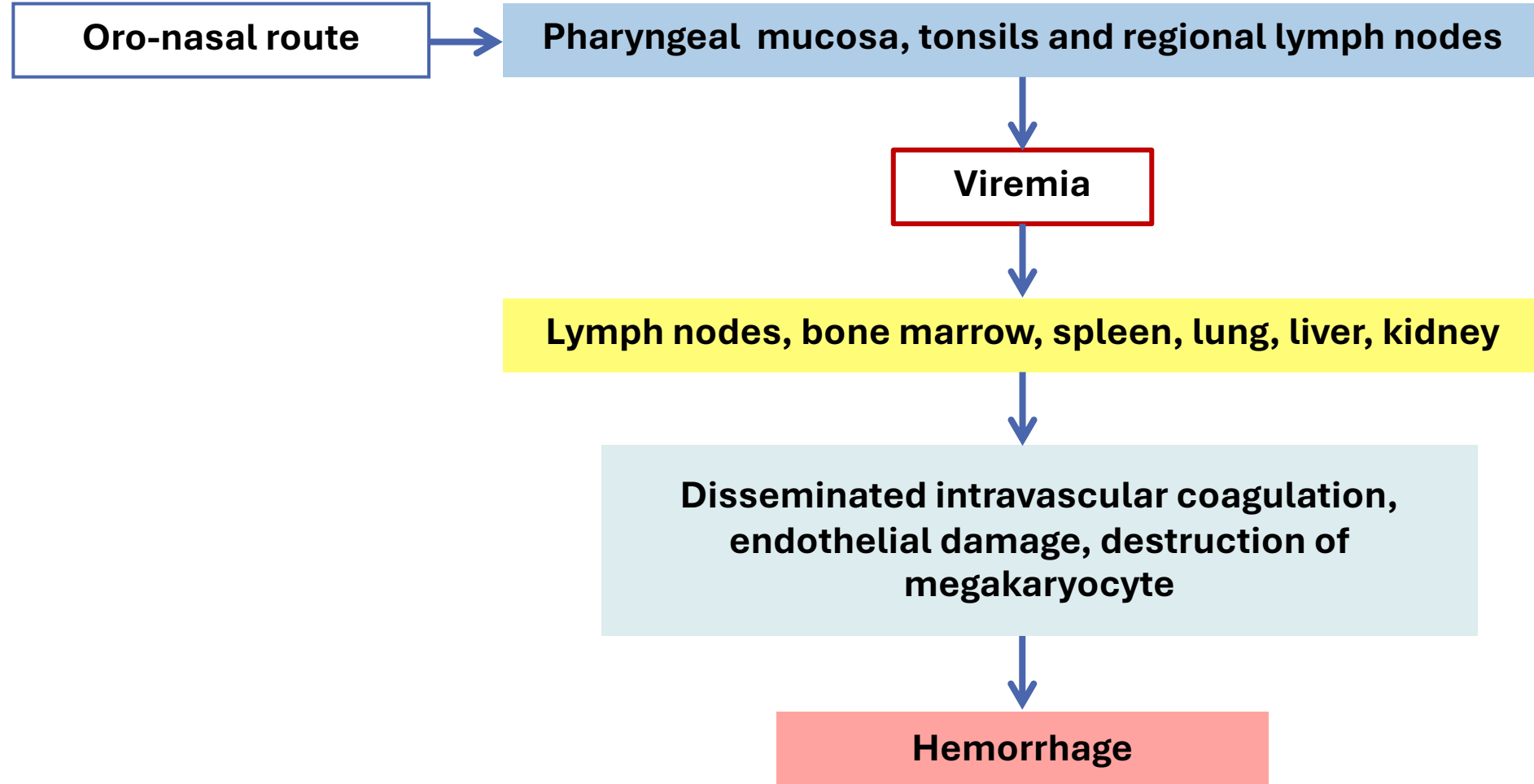
African swine fever

- Importation of wild suidae to Thailand must come from the country/region/zone that free from African swine fever
- African swine fever virus is thermo labile and sensitive to lipid solvents
 - Virus is very resistant to a wide range of pH (several hours at pH 4 or pH 13)
 - Survives for months and even years in refrigerated meat
 - 11 days in feces (room temp.)
 - 1 month in soiled pig pens
 - After 15 weeks in chilled meat
 - 5 - 6 months in processed hams
 - 15 years in frozen carcasses

Transmission



Pathogenesis



Clinical signs

- **Peracute form**
 - Sudden death
 - Swine may die at this stage without other clinical signs
- **Subacute form**
 - Less intense
 - Slight fever
 - Anorexia, weight loss, depress
 - Dying with in 15 - 45 days
 - Mortality 30-70%



Acute form of ASF

- A severe, hemorrhagic disease with high mortality
- Incubation period 5 -15 days
- High fever (40.5 – 42°C)
 - Which persists for about 4 days
 - Starting 1-2 days after the onset of fever
 - Inappetence, diarrhea, incoordination, huddling and prostration
- In some swine there is dyspnea, vomiting, nasal and conjunctival discharge, reddening or cyanosis of the ears and snout, and hemorrhages from the nose and anus
- Pregnant sows often abort
- Mortality is often 100%
 - Domestic swine dying within 1-3 days after the onset of fever



Chronic form of ASF

- Transient/recurrent fever
- Stunting/emaciation
- Pneumonia
- Skin ulcers
- Infected adult warthogs do not develop clinical disease



Diagnosis and control

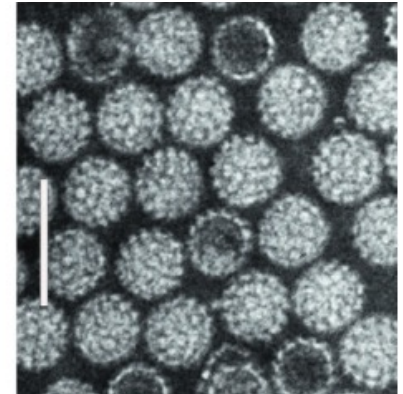
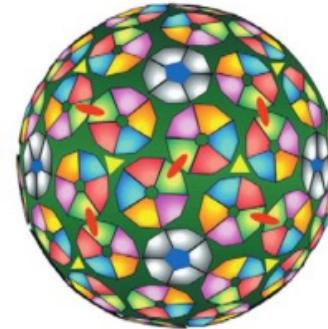
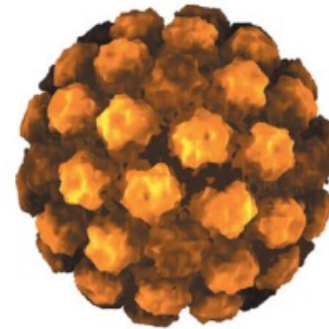
- **Diagnosis**
 - Direct immunofluorescence
 - Hemadsorption test
 - PCR
- **Control**
 - *No vaccine*
 - Free disease country: strict on importation
 - Endemic area:
 - Contact between wild and domestic pig
 - Tick control
 - Feed quality

Family *Papillomaviridae* and Family *Polyomaviridae*

- The family *Papillomaviridae*, like the family *Polyomaviridae*, includes viruses with circular double-stranded DNA genomes
 - The *Papillomaviridae* currently are divided into some 16 genera
 - Alpha-, Beta-, Gamma-, Delta-, Epsilon-, Zeta-, Eta-, Theta-, Iota-, Kappa-, Lambda-, Mupa-, Nupa-, Xipa-, Omikron- and Pipa-papillomavirus

Virion properties

- Virions are non-enveloped, spherical in outline, with icosahedral symmetry
- Virions are 55 nm (*Papillomaviridae*) or 45 nm (*Polyomaviridae*) in diameter
- Genome: single molecule of circular double stranded DNA, 6.8-8.4 kbp (*Papillomaviridae*) or 5 kbp (*Polyomaviridae*) in size
- Members of both families replicate in nucleus





Family *Papillomaviridae*

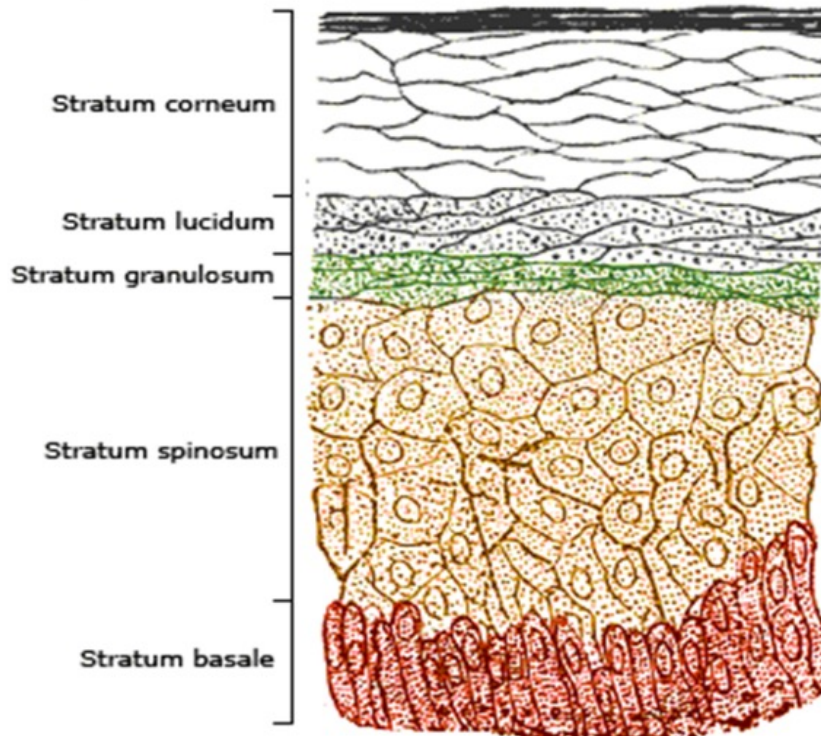
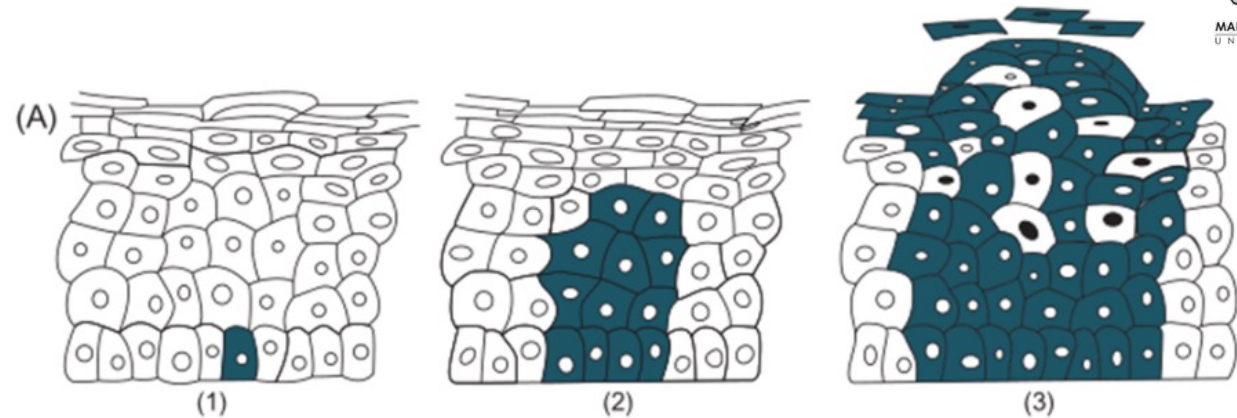
- Papilloma (wart)
 - Benign or malignant
 - Hyperplasia epithelial proliferation of skin and mucous membrane
 - Finger-like projection of proliferating epithelium
- Fibropapilloma
 - Mesenchymal (fibrous) and epithelial tissue
- Papillomaviruses are resistant to diverse environmental insults
 - Infectivity survives lipid solvents and detergents, low pH, and high temperatures

TABLE 11.1 Diseases Caused by Papillomaviruses

Virus	Principal Species Affected	Disease
Bovine papillomaviruses 1–10	Cattle	Cutaneous fibropapilloma and papilloma, teat papilloma, and intestinal papilloma (type 4)
	Horses	Sarcoid (bovine papillomaviruses 1, 2)
Feline papillomavirus(es)	Cat	Cutaneous fibropapilloma (feline sarcoid), plaques and papillomas, squamous cell carcinoma
Ovine papillomavirus	Sheep	Cutaneous fibropapilloma
Equine papillomaviruses 1, 2	Horses	Cutaneous papilloma, aural plaques
Porcine genital papillomavirus	Swine	Cutaneous papilloma
Canine papillomaviruses 1–4	Dogs	Oral papilloma (type 1), endophytic papilloma (type 2), pigmented cutaneous plaques (types 3 and 4)
Deer papillomavirus	Deer	Fibropapilloma, papilloma, fibroma
Cottontail rabbit papillomavirus ^a and rabbit papillomavirus	Rabbits	Cutaneous papilloma (may become malignant)
Fringilla (finch) papillomavirus	Finches	Papilloma
Avian papillomavirus	Parrots	Papilloma

^aAlso called *Shope papillomavirus*.

Pathogenesis



- Papillomavirus infection of keratinocytes
- Primary infection: cell of the stratum germinativum
- Clone of infected cells that spreads laterally
- This is most pronounced in the stratum granulosum
- Virions are shed with exfoliated cells of the stratum corneum

Bovine papillomavirus (BPV)

- BPV 1, 5
 - Cutaneous fibropapilloma
 - Teat frond wart, common cutaneous, rice grain teat papilloma
 - Head, neck, shoulder, omasum, vagina, penis
- BPV 2
 - Benign papilloma -> Invasive carcinoma of urinary bladder
 - Chronic edemic hematuria
- BPV 3, 6
 - No fibroblast proliferation
 - Lesion at epithelium and cutaneous
- BPV 4
 - Benign papilloma -> Invasive carcinoma of alimentary tract
 - Ingestion of fern (*Pteridium aquilinum*)



Transmission and self-limiting

- Transmission
 - Fomites
 - Insemination
 - Housing type
- Self-limiting
 - Autogenous vaccine?

Equine papillomavirus

- Equine papillomatosis
 - Keratinized lesions: around the lips and noses of young horses, but that also can occur on the ears, eyelids, genitalia, and limbs
 - Warts that interfere with the bit, bridle, or other tack can be removed surgically
- Equine sarcoid : this disease have been classified into several types
 - A verrucous (wart-like) type - มักจะไม่มีขน, โตช้า
 - A fibroblastic (proud flesh) type - มีการเจริญของ fibroblast ชั้น intradermis เจริญอย่างรวดเร็วและเป็นแผลหลุม
 - A mixed type - verrucous and fibroblastic
 - A flat type - แบน หนา รอบๆมีขนร่วง





Canine papillomavirus

- Canine oral papillomavirus
 - Lesions within the oral mucosa of young dogs
 - The warts usually begin on the lips, and can spread to the buccal mucosa, tongue, palate, and pharynx
- Self-limiting

Diagnosis and treatment



Diagnosis

- History
- Clinical sign / lesion
- EM
- PCR

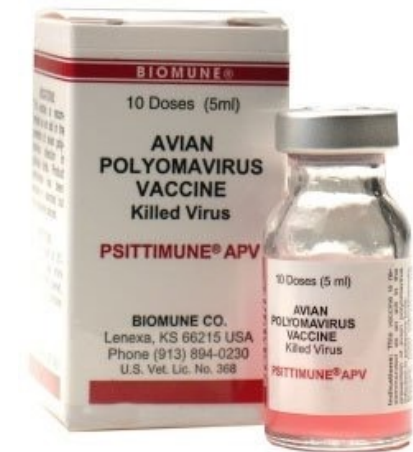
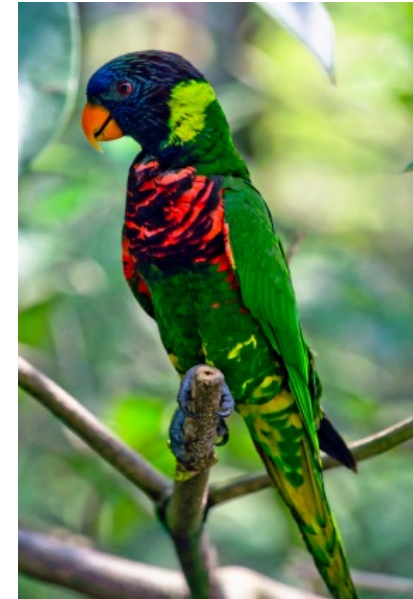


Treatment

- Autogenous vaccine
- Surgery or cryosurgery
- Tincture iodine / glacial acetic acid
- Lithium Antimony thiomate 6%: dose 15 ml
SC q48hr, 4-6 times

Polyomavirus infection

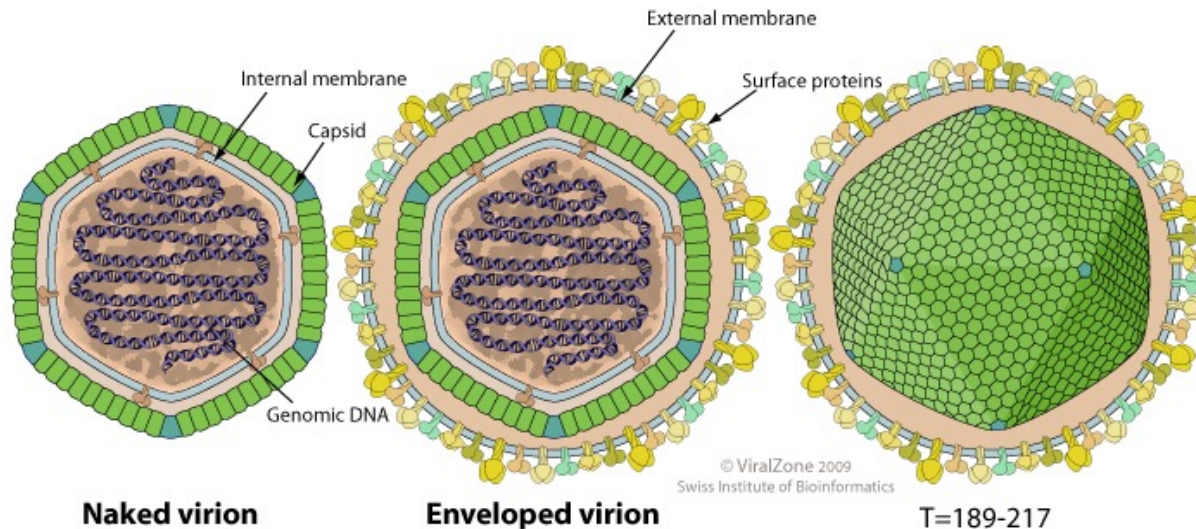
- Host specific
 - Infection in non-specific host -> cancer
 - Example: rodent, rabbit
- **Bovine polyomavirus infection**
 - 60% bovine serum
 - Cancer in immunocompromised rat
- **Budgerigar fledgling disease**
 - USA
 - Acute death, full crop, abdominal distention, reddening of skin
 - Visceral changes: hydropericardium, enlarged heart and liver, congested kidneys
 - Nuclear inclusion bodies, necrotic foci in several organs



Family *Iridoviridae*

- Infect: arthropods, fish, amphibians, and reptiles
- Iridoviruses comprise in 5 genera
 - Genus *Iridovirus*
 - Genus *Chloriridovirus*
 - Genus *Ranavirus**
 - Genus *Megalocytivirus**
 - Genus *Lymphocystivirus*

Virion properties

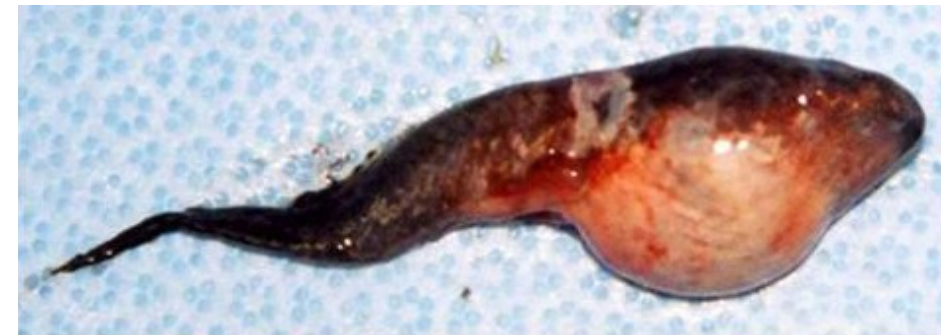


- Icosahedral symmetry, 120-200 nm diameter
- Genome: single linear double-stranded DNA, ranges from 140-200 kbp in size
- A viral envelope is present on virions that bud from infected cells, but is not necessary for infectivity
- Cytoplasmic replication
- Virions exit the cell by budding or cell lysis

Genus *Ranavirus*:

Frog virus 3

- Infected tadpoles
 - Most susceptible
 - Frogs may exhibit localized cutaneous hemorrhage and/or ulceration or severe systemic disease with edema, hemorrhage, and necrosis in numerous organs
 - Death of frog embryo (tadpoles)
 - Tail of tadpole edema
- Subclinical infections: occur in apparently normal wild and captive populations of frogs
- Ranavirus is detected in kidney tissues, including macrophages; which serve as a site for virus persistence



Genus *Megalocytivirus*: Lymphocystis disease virus



Lymphocystis in a walleye

- Benign and self-limiting disease
- Described in a broad range of freshwater and marine fish species
 - Direct contact / Dense population
- Infect and then transform fibroblasts of the skin, gills and internal connective tissues
 - Grey nodules on fins, skin and internal organs
 - Resulting in remarkable hypertrophy of the affected cells
 - Basophilic cytoplasmic inclusion body

References

- Maclachlan, N.J. and Dubovi, E.J. 2011. Fenner's Veterinary Virology. 4th edition. Elsevier Inc., UK. 507p.
- Quinn, P.J., Markey, B.K., Leonard, F.C., FitzPatrick, E.S., Fanning, S. and Hartigan, P.J. 2011. Veterinary Microbiology and Microbial Disease. 2nd edition. Wiley-Blackwell, West Sussex, UK. 912p.

